

Spherical Harmonic Diagonalization of the Different-Ratio Interface BIE

KFBIM notes

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Setting and conventions

Let $\Gamma = S^2$ be the unit sphere. The interior region is the unit ball, the exterior region is its complement, and the normal n points from Ω_{int} to Ω_{ext} . Jumps use the convention from `theory.md`,

$$[u] = \gamma_{\text{int}} u - \gamma_{\text{ext}} u, \quad [\beta \partial_n u] = \beta_{\text{int}} \gamma_{n,\text{int}} u - \beta_{\text{ext}} \gamma_{n,\text{ext}} u.$$

After division by β , each phase solves

$$(-\Delta + \lambda_s^2)u_s = q_s, \quad \lambda_s^2 = \frac{\kappa_s^2}{\beta_s}, \quad s \in \{\text{int}, \text{ext}\}.$$

Here we specialize to

$$\lambda_{\text{int}} = 0, \quad \lambda_{\text{ext}} = \lambda > 0.$$

The formulas below diagonalize the free-space layer potentials on the sphere. They therefore describe the rotationally invariant model problem. If the Green's function is instead the Green's function of a finite rectangular box, as in a KFBIM FFT solve, the box boundary breaks spherical symmetry and the operators are not exactly diagonal in spherical harmonics.

Layer-potential eigenvalues on S^2

Use orthonormal spherical harmonics $Y_{\ell m}$, $\ell = 0, 1, \dots$ and $-\ell \leq m \leq \ell$. For a fixed screening parameter λ_s , write i_ℓ and k_ℓ for the modified spherical Bessel functions of the first and second kind. Define

$$p_\ell^{(s)} = \lambda_s \frac{i'_\ell(\lambda_s)}{i_\ell(\lambda_s)}, \quad q_\ell^{(s)} = \lambda_s \frac{k'_\ell(\lambda_s)}{k_\ell(\lambda_s)}.$$

For $\lambda_s = 0$ these are understood by their Laplace limits,

$$p_\ell^{(0)} = \ell, \quad q_\ell^{(0)} = -(\ell + 1).$$

For density $Y_{\ell m}$, the single-layer potential has boundary value $\mathbf{s}_\ell^{(s)} Y_{\ell m}$, with regular radial factor in the ball and decaying radial factor outside. Since

$$[\partial_n \mathcal{S}_s Y_{\ell m}] = \partial_n (\mathcal{S}_s Y_{\ell m})_{\text{int}} - \partial_n (\mathcal{S}_s Y_{\ell m})_{\text{ext}} = Y_{\ell m},$$

one obtains

$$\boxed{\mathbf{s}_\ell^{(s)} = \frac{1}{p_\ell^{(s)} - q_\ell^{(s)}}.}$$

Therefore

$$S_s Y_{\ell m} = \mathbf{s}_\ell^{(s)} Y_{\ell m}.$$

The averaged normal derivative of the single layer gives the adjoint double-layer operator,

$$K_s^* Y_{\ell m} = \mathbf{k}_\ell^{(s)} Y_{\ell m}, \quad \boxed{\mathbf{k}_\ell^{(s)} = \frac{p_\ell^{(s)} + q_\ell^{(s)}}{2(p_\ell^{(s)} - q_\ell^{(s)})}.$$

On the sphere the double-layer operator has the same eigenvalues,

$$K_s Y_{\ell m} = \mathbf{k}_\ell^{(s)} Y_{\ell m}.$$

For the classical double-layer potential $\mathcal{D}_s$ with $[\mathcal{D}_s Y_{\ell m}] = -Y_{\ell m}$, the interior and exterior traces are

$$\gamma_{\text{int}} \mathcal{D}_s Y_{\ell m} = \left(\mathbf{k}_\ell^{(s)} - \frac{1}{2} \right) Y_{\ell m}, \quad \gamma_{\text{ext}} \mathcal{D}_s Y_{\ell m} = \left(\mathbf{k}_\ell^{(s)} + \frac{1}{2} \right) Y_{\ell m}.$$

Its averaged normal derivative is the hypersingular operator

$$H_s Y_{\ell m} = \mathbf{h}_\ell^{(s)} Y_{\ell m}, \quad \boxed{\mathbf{h}_\ell^{(s)} = \frac{p_\ell^{(s)} q_\ell^{(s)}}{p_\ell^{(s)} - q_\ell^{(s)}}.$$

For the interior Laplace phase, these reduce to

$$\boxed{\mathbf{s}_\ell^{(\text{int})} = \frac{1}{2\ell + 1}, \quad \mathbf{k}_\ell^{(\text{int})} = -\frac{1}{2(2\ell + 1)}, \quad \mathbf{h}_\ell^{(\text{int})} = -\frac{\ell(\ell + 1)}{2\ell + 1}.$$

For the exterior screened phase,

$$\mathbf{s}_\ell^{(\text{ext})} = \frac{1}{p_\ell^{(\lambda)} - q_\ell^{(\lambda)}}, \quad \mathbf{k}_\ell^{(\text{ext})} = \frac{p_\ell^{(\lambda)} + q_\ell^{(\lambda)}}{2(p_\ell^{(\lambda)} - q_\ell^{(\lambda)})}, \quad \mathbf{h}_\ell^{(\text{ext})} = \frac{p_\ell^{(\lambda)} q_\ell^{(\lambda)}}{p_\ell^{(\lambda)} - q_\ell^{(\lambda)}},$$

where $p_\ell^{(\lambda)}$ and $q_\ell^{(\lambda)}$ mean the above logarithmic derivatives evaluated at $\lambda > 0$.

Different-ratio BIE in each spherical harmonic mode

Let a_h and b_h denote the forcing-corrected jumps from Section 6.2 of `theory.md`. If there is no volume forcing, then $a_h = a$ and $b_h = b$. Expand

$$\begin{aligned} \phi &= \sum_{\ell, m} \phi_{\ell m} Y_{\ell m}, & \psi &= \sum_{\ell, m} \psi_{\ell m} Y_{\ell m}, \\ a_h &= \sum_{\ell, m} a_{\ell m} Y_{\ell m}, & b_h &= \sum_{\ell, m} b_{\ell m} Y_{\ell m}. \end{aligned}$$

The Section 6.2 second-kind system is

$$\left[I - \frac{2\beta_{\text{ext}}}{\beta_{\text{int}} + \beta_{\text{ext}}} K_{\text{int}} + \frac{2\beta_{\text{int}}}{\beta_{\text{int}} + \beta_{\text{ext}}} K_{\text{ext}} \right] \phi + (S_{\text{int}} - S_{\text{ext}}) \psi = a_h,$$

$$\frac{4\beta_{\text{int}}\beta_{\text{ext}}}{\beta_{\text{int}} + \beta_{\text{ext}}^2}(H_{\text{ext}} - H_{\text{int}})\phi + \left[I + \frac{2\beta_{\text{int}}}{\beta_{\text{int}} + \beta_{\text{ext}}}K_{\text{int}}^* - \frac{2\beta_{\text{ext}}}{\beta_{\text{int}} + \beta_{\text{ext}}}K_{\text{ext}}^* \right] \psi = \frac{2b_h}{\beta_{\text{int}} + \beta_{\text{ext}}}.$$

Since every operator is diagonal in $Y_{\ell m}$, each mode solves the 2×2 linear system

$$A_\ell \phi_{\ell m} + B_\ell \psi_{\ell m} = a_{\ell m}, \quad C_\ell \phi_{\ell m} + D_\ell \psi_{\ell m} = \frac{2b_{\ell m}}{\beta_{\text{int}} + \beta_{\text{ext}}}.$$

The coefficients are independent of m :

$$A_\ell = 1 - \frac{2\beta_{\text{ext}}}{\beta_{\text{int}} + \beta_{\text{ext}}}k_\ell^{(\text{int})} + \frac{2\beta_{\text{int}}}{\beta_{\text{int}} + \beta_{\text{ext}}}k_\ell^{(\text{ext})},$$

$$B_\ell = s_\ell^{(\text{int})} - s_\ell^{(\text{ext})},$$

$$C_\ell = \frac{4\beta_{\text{int}}\beta_{\text{ext}}}{\beta_{\text{int}} + \beta_{\text{ext}}^2} \left(h_\ell^{(\text{ext})} - h_\ell^{(\text{int})} \right),$$

$$D_\ell = 1 + \frac{2\beta_{\text{int}}}{\beta_{\text{int}} + \beta_{\text{ext}}}k_\ell^{(\text{int})} - \frac{2\beta_{\text{ext}}}{\beta_{\text{int}} + \beta_{\text{ext}}}k_\ell^{(\text{ext})}.$$

Equivalently, with

$$\Delta_\ell = A_\ell D_\ell - B_\ell C_\ell,$$

the modal densities are

$$\phi_{\ell m} = \frac{D_\ell a_{\ell m} - B_\ell \frac{2b_{\ell m}}{\beta_{\text{int}} + \beta_{\text{ext}}}}{\Delta_\ell}, \quad \psi_{\ell m} = \frac{-C_\ell a_{\ell m} + A_\ell \frac{2b_{\ell m}}{\beta_{\text{int}} + \beta_{\text{ext}}}}{\Delta_\ell}.$$

Solvability of every mode

Assume $\beta_{\text{int}} > 0$, $\beta_{\text{ext}} > 0$, and $\lambda > 0$. The modal system is solvable if and only if its homogeneous problem has only the zero solution. Suppose, therefore, that $a_{\ell m} = b_{\ell m} = 0$ and that $(\phi_{\ell m}, \psi_{\ell m})$ solves the homogeneous modal system. Form the corresponding homogeneous layer-potential fields

$$u_{\text{int}} = \mathcal{S}_{\text{int}}\psi - \alpha_{\text{int}}\mathcal{D}_{\text{int}}\phi, \quad u_{\text{ext}} = \mathcal{S}_{\text{ext}}\psi - \alpha_{\text{ext}}\mathcal{D}_{\text{ext}}\phi,$$

where

$$\alpha_{\text{int}} = \frac{2\beta_{\text{ext}}}{\beta_{\text{int}} + \beta_{\text{ext}}}, \quad \alpha_{\text{ext}} = \frac{2\beta_{\text{int}}}{\beta_{\text{int}} + \beta_{\text{ext}}}.$$

The two BIE rows are exactly the interface conditions

$$\gamma_{\text{int}}u_{\text{int}} = \gamma_{\text{ext}}u_{\text{ext}}, \quad \beta_{\text{int}}\gamma_{n,\text{int}}u_{\text{int}} = \beta_{\text{ext}}\gamma_{n,\text{ext}}u_{\text{ext}}.$$

The fields satisfy

$$-\Delta u_{\text{int}} = 0 \quad \text{in } B_1, \quad (-\Delta + \lambda^2)u_{\text{ext}} = 0 \quad \text{in } \mathbb{R}^3 \setminus \overline{B_1},$$

and the exterior field decays at infinity.

Multiplying the interior equation by u_{int} and the exterior equation by u_{ext} , integrating by parts, and using the outward normal conventions gives

$$\begin{aligned} & \beta_{\text{int}} \int_{B_1} |\nabla u_{\text{int}}|^2 dx + \beta_{\text{ext}} \int_{\mathbb{R}^3 \setminus B_1} (|\nabla u_{\text{ext}}|^2 + \lambda^2 |u_{\text{ext}}|^2) dx \\ &= \int_{S^2} (\beta_{\text{int}} u_{\text{int}} \gamma_{n,\text{int}} u_{\text{int}} - \beta_{\text{ext}} u_{\text{ext}} \gamma_{n,\text{ext}} u_{\text{ext}}) dS. \end{aligned}$$

The boundary integral vanishes because the trace and flux are continuous. Hence $u_{\text{ext}} = 0$ and u_{int} is constant. Continuity with $u_{\text{ext}} = 0$ gives $u_{\text{int}} = 0$.

It remains only to exclude a nonzero null representation of the zero field. For one spherical harmonic mode, the zero interior trace gives

$$\mathbf{s}_\ell^{(\text{int})} \psi_{\ell m} - \alpha_{\text{int}} \left(\mathbf{k}_\ell^{(\text{int})} - \frac{1}{2} \right) \phi_{\ell m} = 0,$$

or, using the Laplace eigenvalues,

$$\psi_{\ell m} = -\alpha_{\text{int}}(\ell + 1)\phi_{\ell m}.$$

The zero exterior trace gives

$$\mathbf{s}_\ell^{(\text{ext})} \psi_{\ell m} - \alpha_{\text{ext}} \left(\mathbf{k}_\ell^{(\text{ext})} + \frac{1}{2} \right) \phi_{\ell m} = 0.$$

Since

$$\mathbf{k}_\ell^{(\text{ext})} + \frac{1}{2} = \frac{p_\ell^{(\lambda)}}{p_\ell^{(\lambda)} - q_\ell^{(\lambda)}} = p_\ell^{(\lambda)} \mathbf{s}_\ell^{(\text{ext})},$$

this becomes

$$\psi_{\ell m} = \alpha_{\text{ext}} p_\ell^{(\lambda)} \phi_{\ell m}.$$

Here $p_\ell^{(\lambda)} > 0$, while $\alpha_{\text{int}}, \alpha_{\text{ext}} > 0$ and $\ell + 1 > 0$. The two displayed relations are compatible only when $\phi_{\ell m} = 0$, and then $\psi_{\ell m} = 0$. Thus the homogeneous modal system has no nonzero solution, so

$$\boxed{\Delta_\ell \neq 0 \quad \text{for every } \ell = 0, 1, 2, \dots}$$

Large-mode asymptotics

Let

$$N_\ell = 2\ell + 1.$$

The asymptotics below are for fixed exterior screening parameter $\lambda > 0$ as $\ell \rightarrow \infty$. The regular and decaying logarithmic derivatives satisfy

$$p_\ell^{(\lambda)} = \ell + \frac{\lambda^2}{N_\ell + 2} + O(N_\ell^{-3}), \quad q_\ell^{(\lambda)} = -(\ell + 1) - \frac{\lambda^2}{N_\ell - 2} + O(N_\ell^{-3}).$$

Consequently,

$$\mathbf{s}_\ell^{(\text{ext})} = \frac{1}{N_\ell} - \frac{2\lambda^2}{N_\ell(N_\ell^2 - 4)} + O(N_\ell^{-5}),$$

$$\mathbf{k}_\ell^{(\text{ext})} = -\frac{1}{2N_\ell} - \frac{\lambda^2}{N_\ell(N_\ell^2 - 4)} + O(N_\ell^{-4}),$$

and

$$\mathbf{h}_\ell^{(\text{ext})} - \mathbf{h}_\ell^{(\text{int})} = -\frac{\lambda^2(N_\ell^2 - 3)}{2N_\ell(N_\ell^2 - 4)} + O(N_\ell^{-3}).$$

Substituting these into the modal matrix gives

$$A_\ell = 1 + \frac{\beta_{\text{ext}} - \beta_{\text{int}}}{\beta_{\text{int}} + \beta_{\text{ext}} N_\ell} - \frac{2\beta_{\text{int}}\lambda^2}{\beta_{\text{int}} + \beta_{\text{ext}} N_\ell(N_\ell^2 - 4)} + O(N_\ell^{-4}),$$

$$B_\ell = \frac{2\lambda^2}{N_\ell(N_\ell^2 - 4)} + O(N_\ell^{-5}),$$

$$C_\ell = -\frac{2\beta_{\text{int}}\beta_{\text{ext}}\lambda^2}{\beta_{\text{int}} + \beta_{\text{ext}}^2} \frac{N_\ell^2 - 3}{N_\ell(N_\ell^2 - 4)} + O(N_\ell^{-3}),$$

$$D_\ell = 1 + \frac{\beta_{\text{ext}} - \beta_{\text{int}}}{\beta_{\text{int}} + \beta_{\text{ext}} N_\ell} + \frac{2\beta_{\text{ext}}\lambda^2}{\beta_{\text{int}} + \beta_{\text{ext}} N_\ell(N_\ell^2 - 4)} + O(N_\ell^{-4}).$$

In particular,

$$B_\ell = O(\ell^{-3}), \quad C_\ell = O(\ell^{-1}), \quad A_\ell, D_\ell = 1 + O(\ell^{-1}),$$

and the determinant obeys

$$\Delta_\ell = 1 + \frac{2(\beta_{\text{ext}} - \beta_{\text{int}})}{\beta_{\text{int}} + \beta_{\text{ext}} N_\ell} + O(N_\ell^{-2}).$$

Thus the modal matrices become identity plus a compact lower-order perturbation at high spherical harmonic degree.

Remarks

The leading hypersingular pieces cancel in the same way as in the general smooth-interface derivation. On the sphere this cancellation is visible mode by mode: $H_{\text{ext}} - H_{\text{int}}$ is represented only by the difference of the scalar eigenvalues $\mathbf{h}_\ell^{(\text{ext})}$ and $\mathbf{h}_\ell^{(\text{int})}$, while the prefactor $4\beta_{\text{int}}\beta_{\text{ext}}/(\beta_{\text{int}} + \beta_{\text{ext}}^2)$ is bounded.

The final implementation paragraph in Section 6.2 of `theory.md` uses the KFBIM double-layer jump primitive $[u] = \phi$, which is the opposite of the classical double-layer sign used above. The modal diagonalization is unchanged; only the signs in the double-layer rows should be replaced by the implementation variant written there.